Requirement Analysis

Technology Stack (Architecture and Stack)

Date	02 October 2026
Team ID	SWTID-2026-5170
Project Name	Script-Controlled ACL – Restrict Record Access Based on Custom Script
Maximum Marks	4 Marks

Technical architecture

The solution is built entirely on the ServiceNow platform. Users interact through the ServiceNow web UI; the platform's ACL engine runs server-side JavaScript ACL scripts against the custom table.

Table 1: Components and technologies

S.No	Component	Description	Technology
1	User interface	How users interact with the application	ServiceNow web UI (lists and forms)
2	Application logic-1	Decides whether access is granted	ACL scripts (server-side JavaScript, Glide API: gs, current)
3	Application logic-2	Identity and role information	sys_user, sys_user_role, sys_user_has_role
4	Database	Stores records and rules	ServiceNow built-in database; custom table u_confidential_record; sys_security_acl
5	Cloud database	Database service on cloud	ServiceNow instance database
6	File storage	Not required	N/A
7	External API	Optional testing through REST	ServiceNow REST Table API (also protected by ACLs)
8	Machine learning model	Not required	N/A
9	Infrastructure	Deployment	ServiceNow Personal Developer Instance (cloud)

Table 2: Application characteristics

S.No	Characteristic	Description	Technology
1	Open-source frameworks	None needed; platform-native	ServiceNow platform, server-side JavaScript
2	Security implementations	Role-based access plus script-controlled ACLs	ACLs, roles, security_admin elevation, impersonation tests
3	Scalable architecture	Rules are evaluated per request by the platform	ServiceNow multi-tenant cloud architecture
4	Availability	Hosted by ServiceNow (a developer instance may hibernate when inactive)	ServiceNow cloud
5	Performance	Light scripts, no queries inside ACLs	Field and user checks only